

Peyyala Karthik

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SKILLS

- **Languages:** C/C++, Python
- **Embedded & Electronics:** Microcontrollers, Arduino, ESP32, Sensors, Embedded Systems, PCB Design
- **Simulation & Design:** Proteus, Tinkercad, GitHub
- **Soft Skills:** Problem-Solving, Adaptability, Communication, Teamwork

PROJECTS

2D Memory Controller Using Decoder | Proteus

May' 26

- Designed a decoder-based memory controller to handle memory selection, data storage and output display in a simulated digital system.
- Used a 74139 decoder for memory addressing along with D flip-flops, Write Enable and data switches for storing and controlling data.
- Configured 16 memory locations and used a 7447 decoder with 7-segment displays to represent the selected data.
- Strengthened understanding of memory organization, decoder applications, sequential logic and digital-circuit simulation using Proteus.

Tech stack: Proteus, 74139 Decoder, D Flip-Flops, 7447 Decoder, 7-Segment Displays

Water Level Indicator & Automatic Pump Control | Arduino

Apr' 26

- Built a water-tank monitoring system that measures the water level and provides a visual indication of the measured distance.
- Interfaced an ultrasonic sensor with Arduino to obtain water-level measurements and displayed the readings on an LCD.
- Integrated 1 relay-controlled motor pump to automate the pumping operation based on the system's water-level condition.
- Developed practical understanding of sensor interfacing, LCD display control, relay operation and Arduino-based automation.

Tech stack: Arduino, Ultrasonic Sensor, Relay, LCD Display

Obstacle Avoiding Car | Arduino

Jun' 25

- Developed a small autonomous vehicle that detects obstacles in its path and changes direction without requiring manual control.
- Programmed Arduino to process distance information from the ultrasonic sensor and determine the appropriate movement response.
- Combined 1 ultrasonic sensor and 1 L293D motor driver to implement obstacle detection and differential motor control.
- Learned the fundamentals of embedded control, real-time sensor input, motor interfacing and autonomous vehicle movement.

Tech stack: Arduino, Ultrasonic Sensor, L293D Motor Driver

Automatic Garage Gate Control System | PLC

Dec' 25

- Designed an automated garage-entry and exit sequence in which vehicle-detection sensors initiate the corresponding gate operation.
- Configured the PLC to receive sensor signals and control the opening and closing sequence of the garage gates.
- Implemented 2 gate-control sequences covering both vehicle entry and vehicle exit with timed gate closure
- Gained hands-on understanding of PLC-based sequential control, sensor-triggered automation and industrial control logic.

Tech stack: PLC, Sensors

Word Scramble Puzzle Game | Python

May' 25

- Created a Python-based word puzzle that challenges users to identify words from scrambled characters through an interactive game flow.
- Applied fundamental Python programming concepts to handle word manipulation, user input, game logic and result processing.
- Structured the game around 4 key stages: displaying the scrambled word, accepting the answer, validating the response and progressing through the game.
- Improved programming fundamentals through practical application of conditional logic, data handling, problem-solving and teamwork.

Tech stack: Python

ACTIVITIES AND PARTICIPATION

Aethera-X – 24-Hour Hackathon

Nov' 25

- Participated in a 24-hour hackathon, collaborating with team members to develop and present a solution within a time-constrained environment.
- Gained practical exposure to teamwork, problem-solving, idea development, time management and presenting a technical solution under pressure.

Community Development Project (CDP)

Jul' 26

- Participated as a volunteer in the Community Development Project (CDP), contributing to community-oriented activities as part of the academic program.

- Developed practical experience in teamwork, communication, responsibility and community engagement through participation in the project.

CERTIFICATES

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| • Cyber-Physical Systems: Sensors and PLC Interfacing – Centre for Professional Enhancement, LPU Link | Nov' 25 |
| • Basics of Electrical Safety – UniAthena Link | Dec' 25 |

EDUCATION

Lovely Professional University	Punjab, India
<i>Bachelor of Technology - Electronics and communication Engineering; CGPA: 7.10</i>	<i>Aug' 25 - Present</i>
Tirumala Junior College	Rajahmundry, Andhra
Pradesh	
<i>Intermediate; Percentage: 94.4%</i>	<i>Jun' 23 – Apr' 25</i>
Bhashyam English Medium High School	Rajahmundry, Andhra
Pradesh	
<i>Matriculation; Percentage: 85%</i>	<i>Jun' 22 - Apr' 23</i>